# POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Basic Electrical Engineering (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

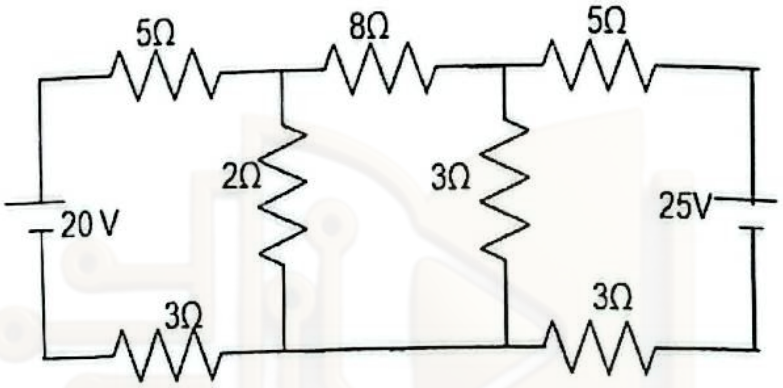
*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

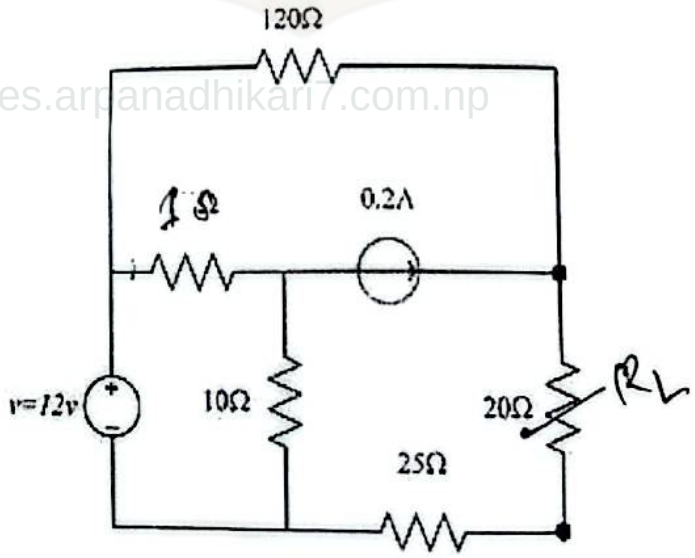
***Attempt all the questions.***

1. a) Describe in brief the generation, transmission, distribution and consumption of electrical energy. 7

   b) Find the current across 8Ω resistor using nodal analysis.




2. a) What should be the value of load resistance in place of 20 Ω such that maximum power will transferred to it. Also find the maximum power transferred to it. 8



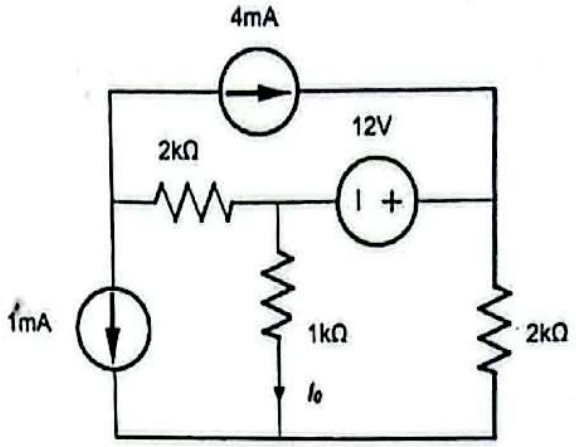

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**OR**

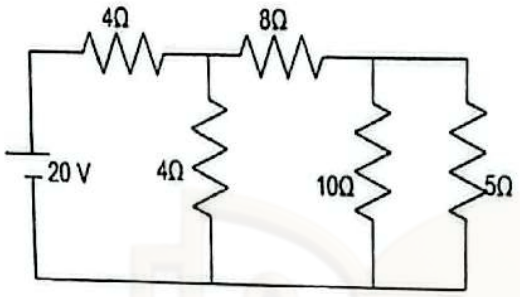


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Find current through 1KΩ using Superposition Theorem.




b) Calculate the current across 5Ω resistor using Norton's theorem. 7




3. a) Find the average value, effective value and form factor for a sinusoidal voltage waveform. 7

b) Two circuits A and B are connected in parallel across a 200V, 50 Hz supply mains. Circuit A consists of a resistance of 100Ω and inductance of 0.12H in series, circuit B consists of a resistance of 20Ω in series with capacitor of 40μF. Calculate: 8

i. current in each branch
ii. Source current
iii. circuit power factor
iv. Real power, reactive power and apparent power.

Also draw phasor diagram.

4. a) A series R-L-C circuit has 10Ω resistance, 0.01H inductance, and 100μF capacitor. A 100V supply with variable frequency is applied to this circuit and tuned until resonance occurs. Find the resonant frequency, bandwidth, and quality factor, current drawn by the circuit, and voltage across inductor at resonance condition. 7

**OR**

Derive the necessary expression for resonant frequency in RLC series circuit.



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b) Three equal impedances having resistance 15 Ω and inductive resistance 12 Ω are connected in delta connected system. Find; 8

i. Phase and line current
ii. Power factor
iii. Power consumed

5. a) Explain how three phase power is measured using two wattmeter in star and delta connected system. 7

b) A 20 KVA, 200/200 V, 50 Hz single phase transformer has the following test results. 8

| | Volt | Ampere | Watt |
|---|---|---|---|
| Open circuit test with H.V. open | 200 | 4 | 120 |
| Short circuit test with L.V. shorted | 60 | 10 | 300 |

Calculate the parameter of equivalent circuit of a transformer referred to H.V. side.

6. a) The power input to a 230 volts DC shunt motor is 8.477 kW. The field resistance is 230 ohm and armature resistance is 0.28 ohm. Find the input current, armature current and back e.m.f. 7

b) Explain the principle of operation of three phase induction motor. 8

7. Write short notes on: **(Any two)** 2×5

a) Kirchhoff's law
b) Power factor and its significance
c) Transformer Losses

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